Introduction

Training System

- **This training is totally for free**
- **First session**.
- Practice session.
- Online contest.

Training outcomes

- Thinking techniques.
- Learn how to learn.
- learn how to solve problems.
- learn how to debug your code.
- Clean code.
- Team work.
- Soft Skills (especially communications).

Training Content

- Intro to computer – problem solving.
- Data Types.
- Conditions.
- Loops.
- Arrays.
- Strings.
- Functions.
- Math.

Don’t forget Reference*

Our content for today :

- What is the computer ?
- Programming & Programming Languages.
- What is the compiler ?
- Online Judges.
- Problem Solving.
- Your first Program.
- Variables.
- Data Types.
- Cin & Cout.

What is the computer ??!

- A **computer** is a programmable device that can store, retrieve, and process data.



Programming & Languages

- **Computer programming** is a way of giving computers instructions about what they should do next .

- **Programming Languages** is a formal **language**, which comprises a set of instructions that produce various kinds of output,

  like (C++, JAVA, PYTHON, ROBY,....., etc).

In this course we will use C++ language.

What is the compiler ?!

- A compiler is a program that translates a high level programming language (called source code) into machine language (the target language).
- Machine language is a sequence of 0's and 1's that the machine (computer) understands and can interpret into instructions.



Compilers Support C++



Code::Blocks





Visual Studio

Online Judges

- An Online Judge system is a web software for compiling, executing and evaluating programs submitted by users.









Let’s Try it

Expert

IcpcSVU

Qina, Egypt
From Fci south valley university

Contest rating: **1790** (max. **expert, 1790**)

Contribution: 0

Friend of: 113 users

My friends

Change settings

icpcsvuqena@gmail.com (not visible)

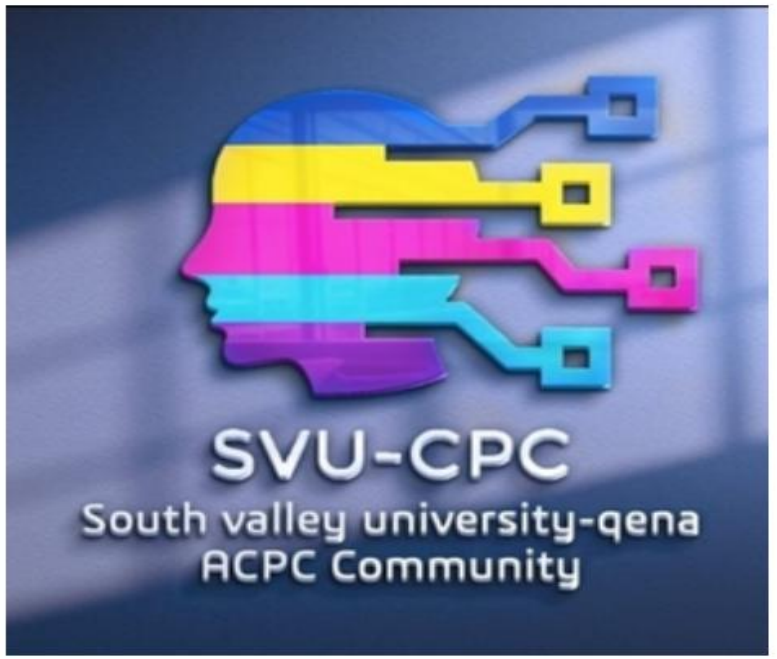
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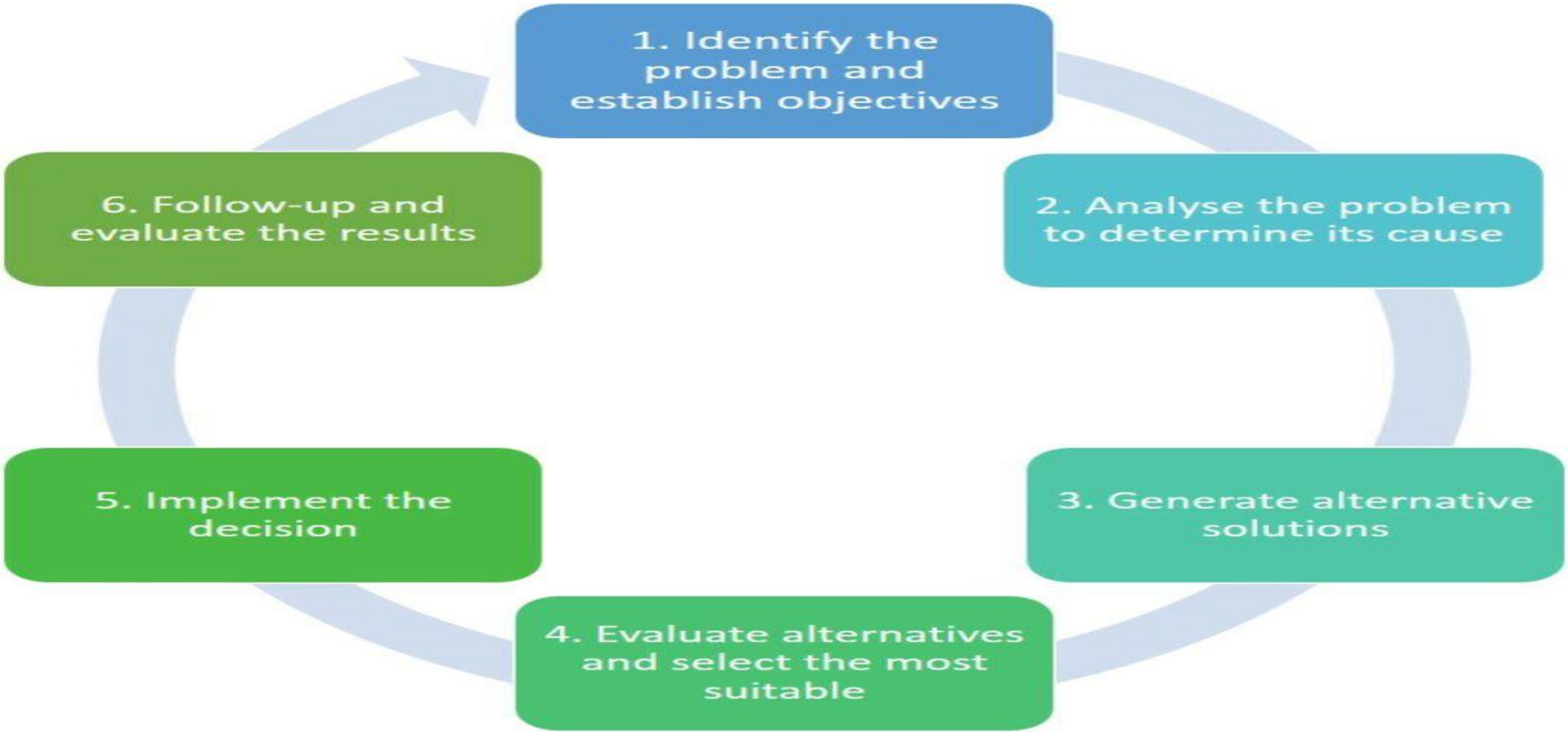
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Problem Solving



Algorithm

Algorithm :- a step-by-step procedure or set of instructions for solving a problem or accomplishing a task.



Algorithm

2 + 2 + 2 = ??



Algorithm

2 + 2 + 2 = 6



Algorithm

How about adding 2 more ?

2 + 2 + 2 = 6



Algorithm

2

$2 + 2 + 2 = 6$

$6 + 2 = 8$

$2 + 2 + 2 + 2 = 8$



Algorithm

2

$2 + 2 + 2 = 6$

✓ $6 + 2 = 8$

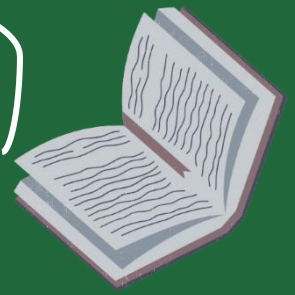
$2 + 2 + 2 + 2 = 8$



Algorithm

We have a problem









Algorithm

4096

In the worst case



One page per step

Algorithm

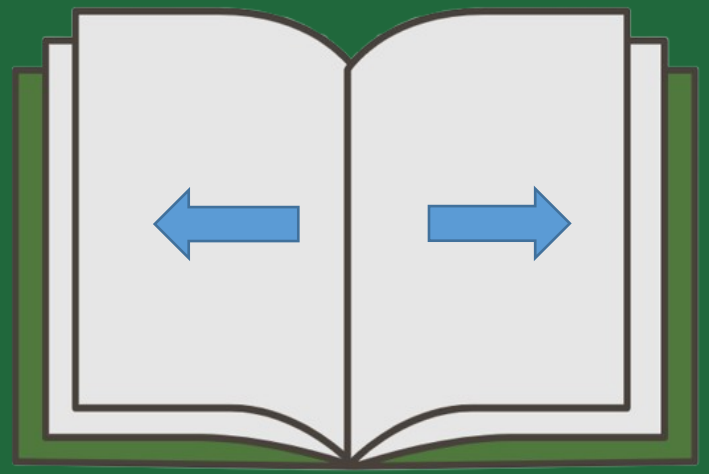
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In the worst case

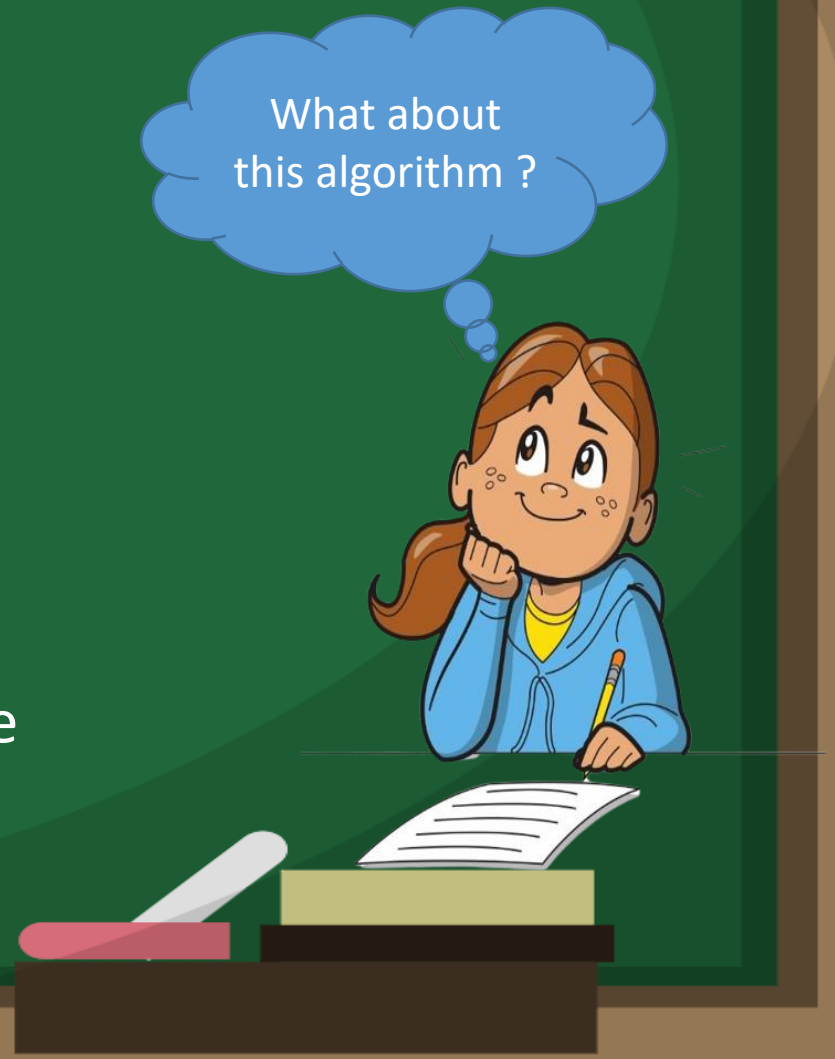


Two page per step

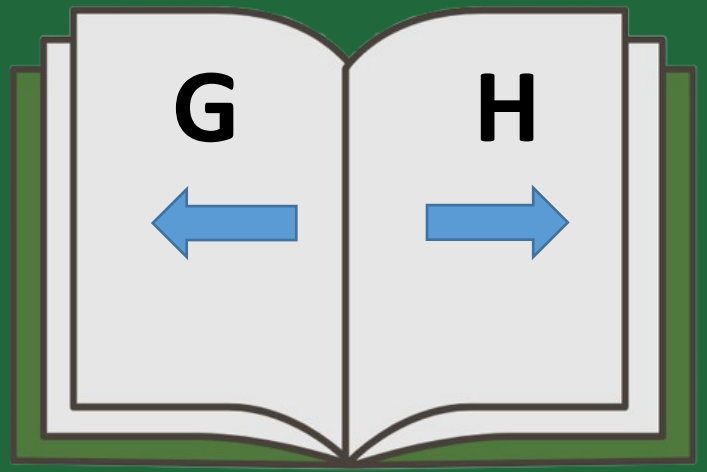
Algorithm



Open the book from the middle



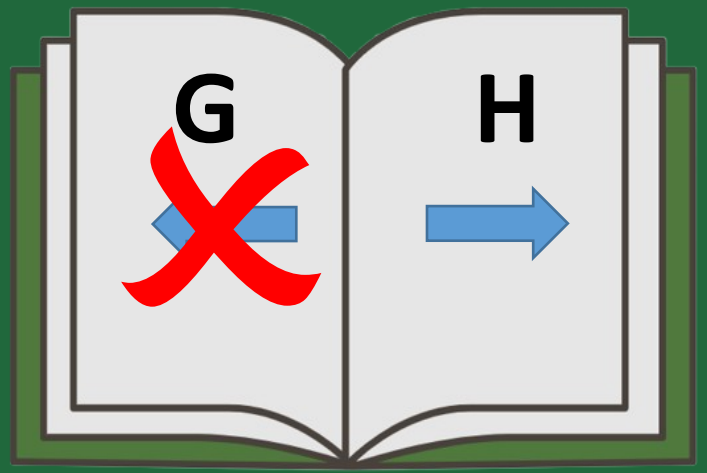
Algorithm





Open the book from the middle

Algorithm

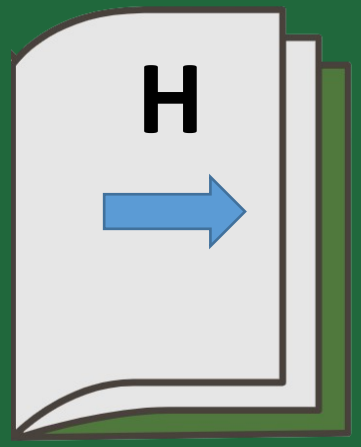


O



We don’t need the left side

Algorithm

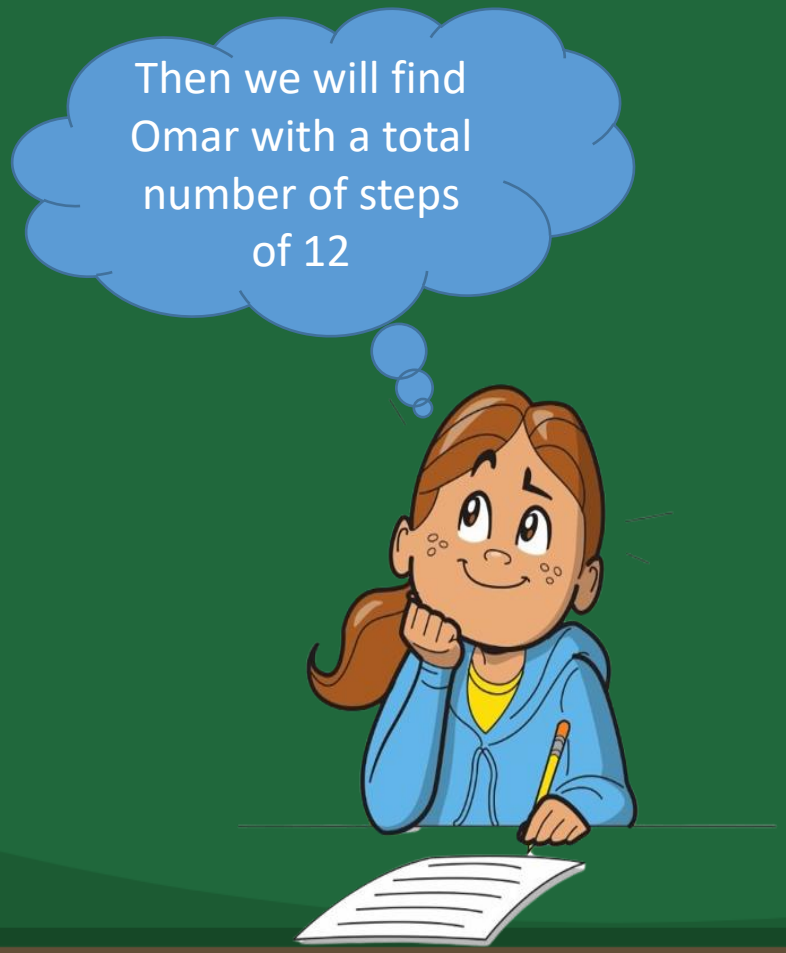


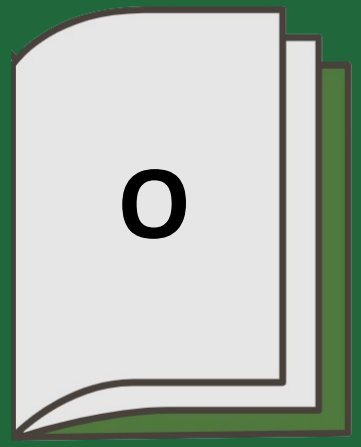
O



We don’t need the left side

Algorithm





Learn problem solving
then go to try any track you want







It’s time to your first program



```cpp
#include <iostream>
using namespace std;
int main()
{
    cout<<"Hello Wrold!"<<endl;
    return 0;
}
```

Structure of the program

#include <iostream> (input output stream)

- Known as header **iostream**, that allows to perform standard input and output operations.

int main ()

- A function is a group of code statements which are given a name: in this case, this gives the name "**main**" to the group of code statements that follow.

- std::cout << "Hello World!";

This statement has three parts:

- -First, std::cout, which identifies the standard character output device (usually, this is the computer screen).
- -Second, the insertion operator (<<), which indicates that what follows is inserted into std::cout.
- -Finally, a sentence within quotes ("Hello world!"), is the content inserted into the standard output.

(;) every statement in c++ end with **semicolon** .

Error

```
#include <iostream>
using namespace std;
int main()
{
    cout<<"I am Omar"
    return 0;
}
```

Correct

```
#include <iostream>
using namespace std;
int main()
{
    cout<<"I am Omar";
    return 0;
}
```

Error



```cpp
#include <iostream>
using namespace std;
int main()
{
    cout<<"I am Omar;
    return 0;
}
```

Correct



```cpp
#include <iostream>
using namespace std;
int main()
{
    cout<<"I am Omar";
    return 0;
}
```

Error

```cpp
#include <iostream>
using namespace std;
int main()
{
    cout<<"Maher is a "< <"Student";
    return 0;
}
```

Correct

```cpp
#include <iostream>
using namespace std;
int main()
{
    cout<<"Maher is a "<<"Student";
    return 0;
}
```

Error

```
1  #include <iostream>
2  using namespace std;
3  int main()
4
5      cout<<"Hello Wrold!"<<endl;
6      return 0;
```

Correct

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout<<"Hello Wrold!"<<endl;
6      return 0;
7  }
```

Error

```
using namespace std;
int main()
{
    cout<<"Hello Wrold!"<<endl;
    return 0;
}
```

Correct

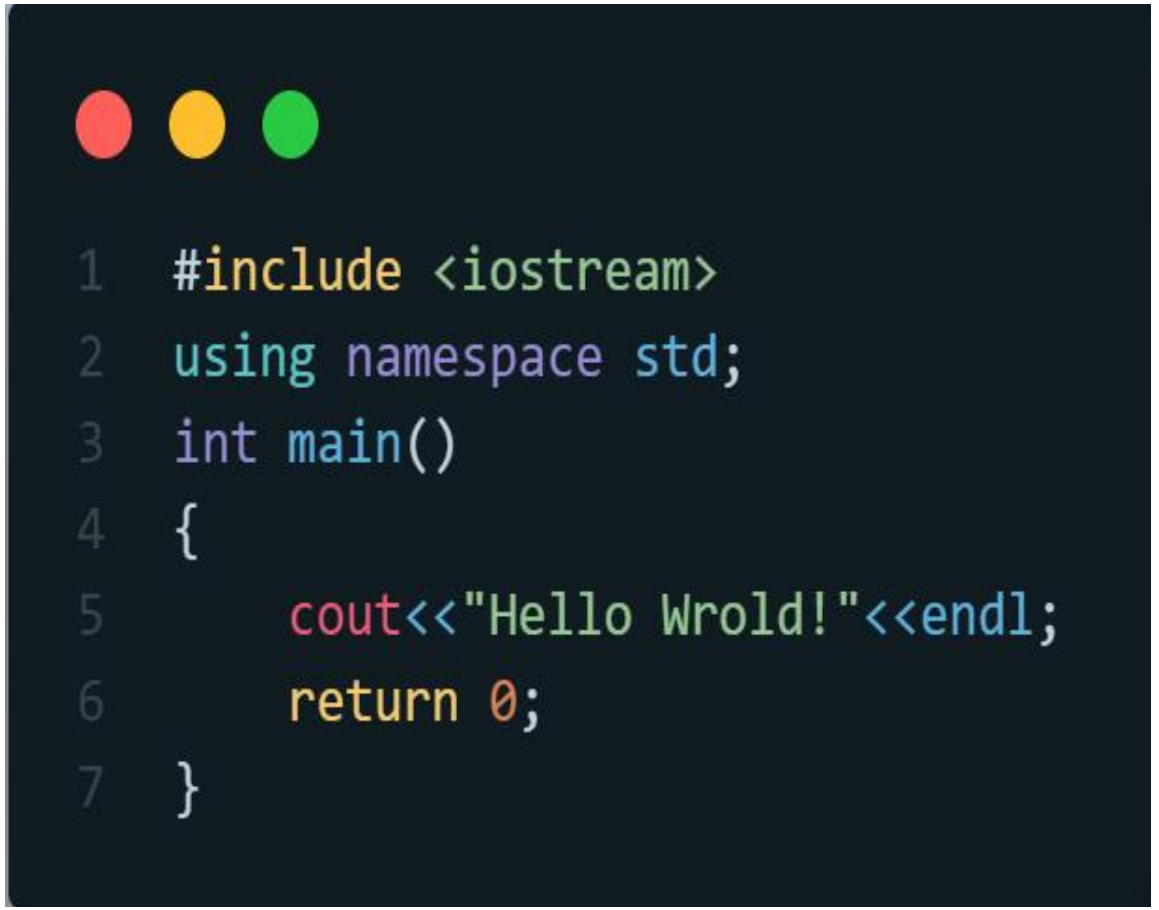
```
#include <iostream>
using namespace std;
int main()
{
    cout<<"Hello Wrold!"<<endl;
    return 0;
}
```

Error



```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout<<Hello Wrold<<endl;
6      return 0;
7  }
```

Correct



```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout<<"Hello Wrold!"<<endl;
6      return 0;
7  }
```

Find all errors and fix them

```
#include <iostream
using namespace STD
cout<<"work smart not hard\n";
int maIN()
{
cout<<"Children must be taught how to think, not what to think"

cout << "We worry about what a child will become "tomorrow", yet we forget that he is someone today\n";

cout<<"Children are not things to be molded"<", but are people to be unfolded\n";

cout <<< ""It is easier to build strong children than to repair broken men""<<"\n";

cout>>"Children need modles rather than critics\n";

out<<"Children have never been very good at listening to their elders, but they have never failed to imitate them";

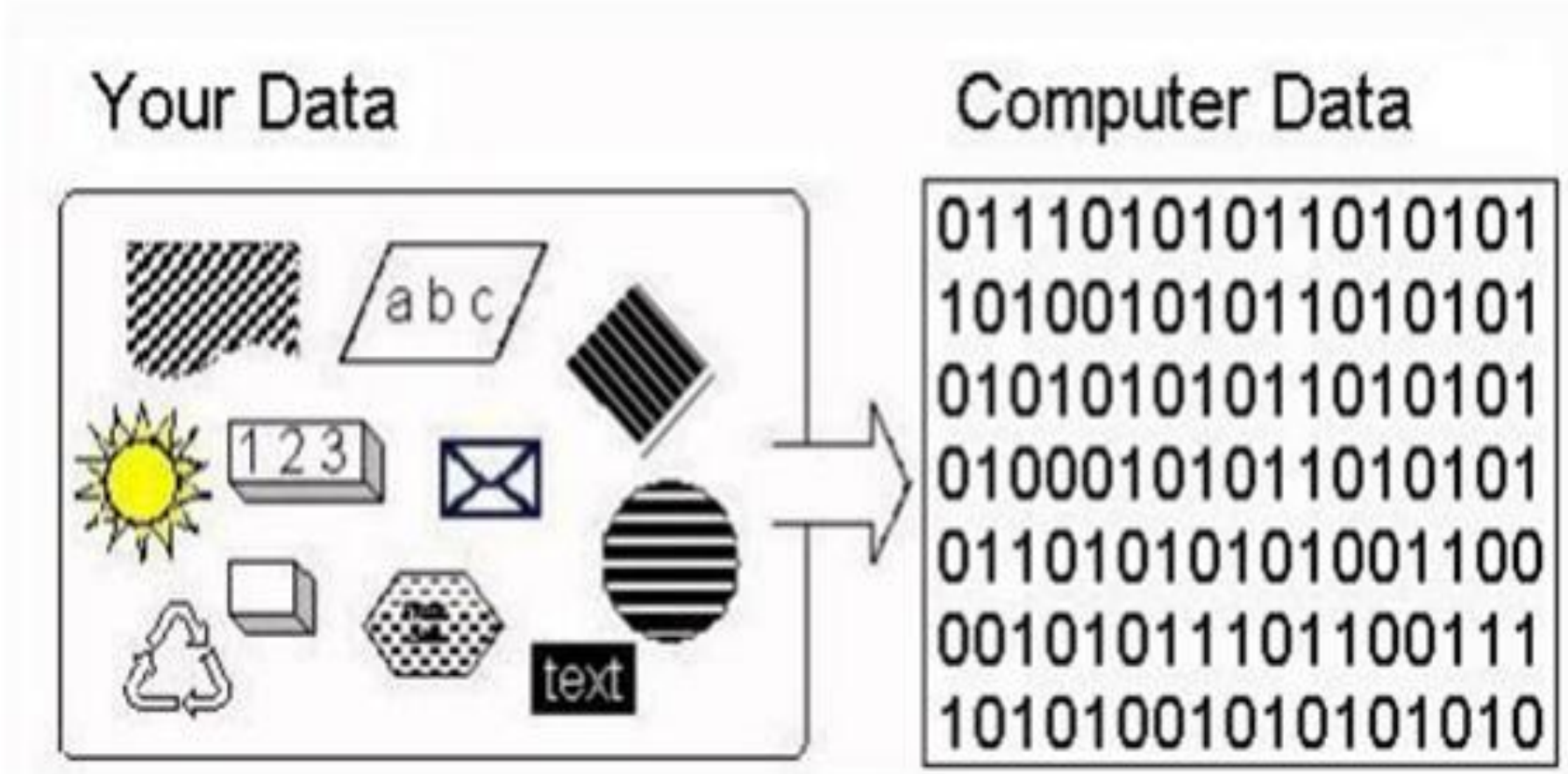
cout<"Children are our most valuable resource\n";

return;
```


BREAK
TIME !!

Data Types and Variables

Binary system



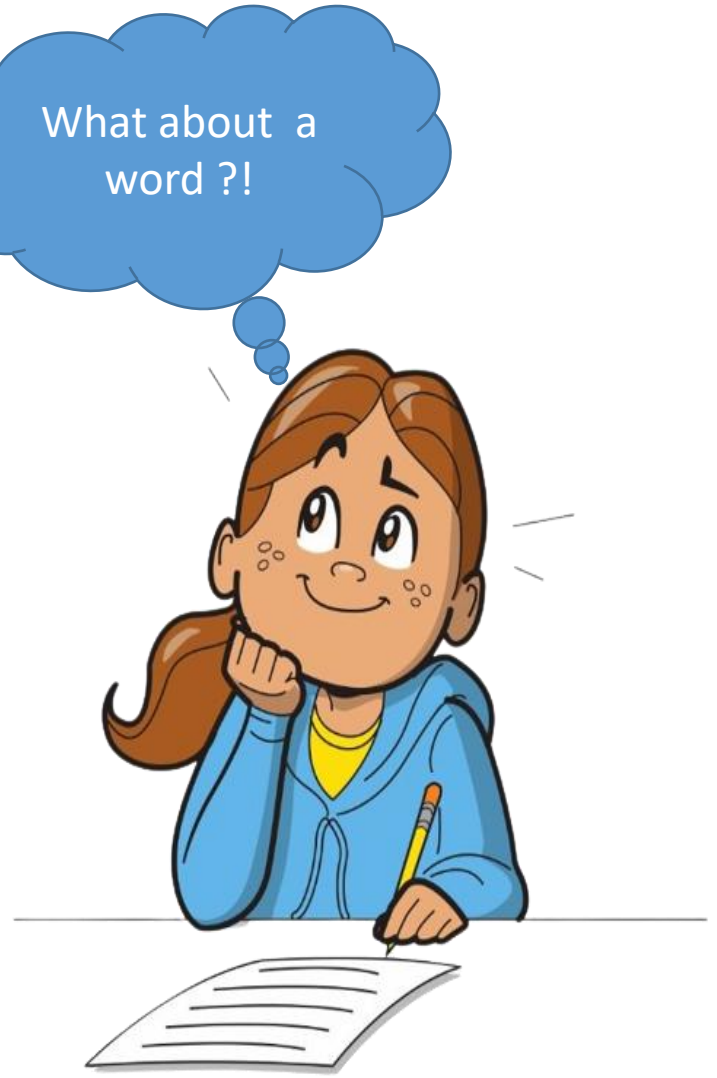
Variables

- Used to store values in memory.
- When naming a variable you should consider the following:
  1. A variable name is a sequence of one or more letters (a-z) (A-Z), digits (0-9) or underscore characters _ .
  2. Neither spaces nor punctuation marks or symbols can be part of a variable name.
  3. In addition, variable names always have to begin with a letter.
  4. Another rule that you have to consider when naming a variable is that they cannot match any keyword of the C++ language nor your compiler's specific ones.

Data Types

KEYWORD	SIZE (in bytes)
int	4
char	1
float	4
double	8
long long	8
boolean	1

1 Byte = 8 Bits



Variable size limit?

Data Type	Size	Range
int	4 bytes	-2^31 -> (2*31) - 1
long long	8 bytes	-2^63 -> 2*63 - 1
float	4 bytes	Precision 7 Decimal Points
double	8 bytes	Precision 15 Decimal Points
char	1 byte	-128 -> 127
bool	1 byte	false, true

String

- A sequence of characters.

Example

```
string name = “ICPCSVU”;
string s = “*m*7&@vyvy”;
```

Variable declarations and initializations, assignment operator =

To create a variable in C++, we do a process called **Variable Declaration**

```
//Data_Type   Variable_Name;
  int x;
```

You can also declare multiple variables of the same type in a single line

```
int x, y, z;
```

Now we want to put some value in that variable, this is called **Variable Assignment**

```
  int x;

// Variable_Name =  Value;
  x = 5;
```

We can even combine the two processes in one called **Variable Initialization**

Note: copy initialization is the most common way

```
//Data_Type    Variable_name = initial value;
  int x = 5; (This is called copy initialization)
  int x(5); (This is called direct initialization)
  int x{5}; (This is called uniform initialization)
```

Uninitialized variables (Garbage!)

When a variable is declared, it's given what was previously stored in its memory location.
It can be something like `5277592` or `4309726`. You can never know.

```
int x;
cout << x; //who knows what we'll get!
```

Naming convention

When naming a variable you should consider the following:

a. A variable name is a sequence of one or more letters (a-z) (A-Z), digits (0-9) or underscore characters _ .

b. Neither spaces nor punctuation marks or symbols can be part of a variable name.

c. In addition, variable names always have to begin with a letter.

d. Another rule that you have to consider when naming a variable is that they cannot match any keyword of the C++ language nor your compiler's specific ones.

- `int 1stNum;  //invalid`
- `int my variable name;  //invalid`
- `int my@variable#name;  //invalid`

- `int myVariableName;  //correct`
- `int Myvariable_name;  //valid but not preferred`
- `int my_variable_name;  //correct`
- `int variable;  //correct`
- `int variable1; //correct`

Example

```
#include <iostream>
using namespace std;
int main()
{
   int x;  // Declaration
   x = 10; // Initialization
   int y = 5; // Declaration and Initialization
   float f; // Declaration
   f = 3.14;  // Initialization
   char c = 'h'; // Declaration and Initialization
   bool state = true; // Declaration and Initialization
   string name = "Ahmed";
   cout << "x = " << x <<endl;
   cout << "y = " << y << endl;
   cout << "f = " << f << endl;
   cout << "c = " << c << endl;
   cout << "state = " << state << endl;
   cout << "name = " << name << endl;

   return 0;
}
```

Operators

Operator	Use	Example	Result
+	To add two numbers	i=3+2	5
-	For subtraction	i=3-2	1
*	For multiplication	i=3*2	6
/	For division	i=3/2	1
%	Modular division (Reminder after division)	i=10%3	1

- Int / int = int
- Int /float = float
- float/ int = float
- int * int = int
- long long * int = long long
- long long * double = double

Compound assignment operators

```
int x = 5;
x -= 2;           //equivalent to x = x - 2
                  //X equals to 3
int y = 4;
y *= x;           //equivalent to y = y*x
                  // y equals 12

y /= 2;           //equivalent to y = y/2
                  // y equals 6
```

Important : (**+=**, **-=**, ***=**, **/=**, **%=**)

Increment & decrement operators

```
int x = 5;
cout << x++;          //means print the value of x (5) to the
console then increment it
                    //X equals to 6, the console prints 5
int y = 4;
cout << --y;          //means decrement the value of y(4) then
print it to the console      // y equals to 3, the console
prints 3
```

Output?

```cpp
#include <iostream>
using namespace std;
int main()
{
    int x = 5;
    cout << ++x << " " << x++ << ' ';
    x++;
    cout << x;
    return 0;
}
```

Formula : a % b = a – (a/b) *b ;

EX : int x = 11 % 3 = 11 - (11 / 3) * 3 = 11 - 3 * 3 = 11 - 9 = 2;

- **Used to :**
- Divisibility.
- Even Odd.
- Last Digit.

Not Work on doubles.

Comments

1. Line comment

```cpp
#include <iostream>
using namespace std;
int main()
{
    cout << "Hello world!" << endl; // print Hello world
    return 0;
}
```

2. Block comment

```cpp
#include <iostream>
using namespace std;
int main()
{
    cout << "Hello world!" << endl;
    /*
        this program will print Hello world
    */
    return 0;
}
```

Standard Input (Cin)

```
int x; // variable of type integer
cin >> x; //
```

```
char x, y, z ; //variables of type character
cin >> x >> y >> z ; //the first character the user enters will be x , the second one y and so on
```

*But take care, the input stops by a space or a new line, which means:
If for example you have a variable x of type int and you want x to have a value = 123
But then while you enter this value you typed : 12 -> space/enter -> 3
Your input stops at the space/enter , so x now has a value = 12 **not** 123.

OUTPUT

- cout : for output the result of the program.

  cout << variable_name;

  int x; cin >> x; cout << x;

  cout << “This is the first session”;

- Another output:

  \n : new line.

  \t : tab.

  \? : question mark.

  \\ : backslash.

Quizz

- Write a program to calculate the sum, multiplication, division, remainder of two given numbers.
- **Input** : 2 7
- **Output** :

```
2 7
Sum = 9
Multiplication = 14
Divsion = 0
Remainder of Divsion = 2

Process returned 0 (0x0)   execution time : 12.434 s
Press any key to continue.
```

Any Questions ???



Your first task

- Register in Codeforces Website.
- Have Code blocks or any compiler.
- Contact with your Mentor.
- Open sheet every day and solve.
- Have your Reference .

It’s enough !!!
Thanks